Train, Validation, Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Train, Validation, Test

> Prepare Train, Validation, Test: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Select a model using a declared **validation** rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate MAE MSE and RMSE

Reference example, not the only acceptable wording: Teacher explanation: (1+2+3)/3=2 cm.

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Concept to explain

Training fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.

Term	Meaning
training set	data used to fit parameters
validation set	data used to choose settings
test set	data used for final estimate
split strategy	rule assigning examples to sets

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **validation** and model selection. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Training rows fitted parameters; separate **validation** rows compared performance.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain train, **validation**, test using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Train, Validation, Test using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Assign every modeling decision and dataset use to the correct split.

MODEL IDEA **Training** fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Fit two candidate trees on **training** rows. Choose depth and threshold using **validation** metrics and error slices.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Fit two candidate trees on **training** rows.

Step 2

STEP 2 Choose depth and threshold using **validation** metrics and error slices.

Step 3

STEP 3 Freeze pre**process**ing, features, model, and decision threshold.

Step 4

STEP 4 Evaluate once on test; do not return to tune and still call it final.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Split 100 independent leaves into 60 train,20 **validation**,20 test.

The teaching case

Explain roles and pre**process**ing.

Exact explanation to model

Why the result follows

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

Explain the added worked example

Explain the added worked example · 1

Fitting and selecting answer different questions. **Training** adjusts parameters. **Validation** helps choose settings or model families. A final test estimates the performance of the frozen development **process** on held-out examples.

Keep related records in the same split when separate rows could reveal the same source.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **validation** and model selection.

Three candidates have **training** MAE A=1, B=0, C=2 and **validation** MAE A=3, B=5, C=3. The declared rule is smallest **validation** MAE, then simpler model, with A simpler than C. Choose A.

B fits **training** perfectly but does worse on **validation**. The final test remains unopened.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Validation	Development evidence used to choose between options
Hyperparameter	A model setting chosen outside its fitted parameters
Selection rule	A declared method for choosing an option
Test set	Held-out evidence reserved for a final evaluation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Validation and model selection at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training	Fit each candidate	No test labels
Validation	Choose A by rule	Record every attempt
Final test	Frozen model only	Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

Where should parameter fitting, setting selection and final reporting occur? Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train, Validation, Test

Assign a separate job to **training**, **validation** and final test **evidence**.

The facts we will use

Supplied facts

Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/24

Train, Validation, Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit on training

Both candidates are built using training data; B fits it more closely.



2. Select on validation

Choose A because2 cm is below3 cm.



3. Reserve final test

Freeze A before scoring final cases; do not select again using their labels.

CHANGE ONE THING

A and B both have validation MAE2 cm; A is simpler under a predeclared tie rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit on training	Both candidates are built using training data; B fits it more closely.
2. Select on validation	Choose A because2 cm is below3 cm.
3. Reserve final test	Freeze A before scoring final cases; do not select again using their labels.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

Change one thing in the pictured case

Change one thing

Change: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Predicted result

Prediction: Select A using the predeclared simpler-model tie-break.

Why the result follows

Reason: The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Train, Validation, Test, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Split 100 independent leaves into 60 train,20 **validation**,20 test.

2. State the decision to explain

Write the question to explain: Explain roles and pre**process**ing.

3. Reveal the worked reasoning

Reveal after **prediction**s: Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

Hold back the independent task

Keep for an independent attempt: Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, **validation** and test?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the

> result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

Exit response: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Which data fitted the line last week? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes
>
> **Teacher action:** Read the opening and model the worked evidence. Ask students to identify validation in context.

> **Stage:** 18 to 32 minutes
>
> **Teacher action:** Complete W 1 and W 2 with support: choose a candidate; assign data roles. Require the evidence, not only the final response.

> **Stage:** 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between validation and hyperparameter. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: A team creates D after reading validation mistakes. What changes in the record?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Select a model using a declared validation rule; Keep training, validation and test roles separate. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Where should parameter **fitting**, setting selection and final reporting occur?

Reasoning to listen for

Teacher check: Teacher explanation: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why does B’s zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Model selection is based on held-out **validation** error here. Lower **training** error can reflect **fitting** the **training** details rather than better new-case **prediction**.

Changed-case check: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

Changed-case reasoning: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Train, Validation, Test investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Role Assignment: sort twenty actions into train/**validation**/test, then group related examples before splitting.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Three random folders are enough even when photos of one plant appear in each.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

A stronger claim makes a stronger project: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Choose P with **validation** MAE 4; Q loses with 6.

Keep the test unopened until settings and reporting rules are frozen.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Train, Validation, Test questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Train fits parameters and learned transforms; **validation** chooses settings; final test

evaluates the selected system. Group related records first.

Independent question

Independent question · withhold the key

Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, **validation** and test?

Expected independent answer

Teacher key · Expected independent answer

A is **training**, B is **validation** and C is final test. Keep C separate from **fitting** and model selection.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “**Validation** and final test data have interchangeable purposes during development.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Image counts are 54,18,18. Fit the mean on **training** images only, then apply the fixed transform to **validation** and test. **Validation** chooses settings; final test evaluates the selected system.

Do not compute the mean from all 90 images.

Explanations for the additional workbook examples

W1 Choose a candidate

Workbook question · W1

Apply the worked **selection rule**. Explain why B is not chosen.

Teacher reasoning · W1

Teacher explanation: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

W2 Assign data roles

Workbook question · W2

Where should parameter **fitting**, setting selection and final reporting occur?

Teacher reasoning · W2

Teacher explanation: Fit on **training**, select on **validation**, evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

W3 Record another attempt

Workbook question · W3

A team creates D after reading **validation** mistakes. What changes in the record?

Teacher reasoning · W3

Teacher explanation: Add D and its motivation to development history. **Validation** influenced D and is not untouched test **evidence**.

W4 Apply independently

Workbook question · W4

Training errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

Teacher reasoning · W4

Teacher explanation: Choose P with **validation** MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

W5 Limit a claim

Workbook question · W5

Why is the winning **validation** score not a guaranteed future score?

Teacher reasoning · W5

Teacher explanation: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

Assess independent transfer

Independent question · withhold the key

Selected case: **Training** errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Choose P with **validation** MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Train, Validation, Test to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 25: Metrics Match Consequences. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **validation** and **hyperparameter** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Choose P with **validation** MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen. Fresh retry: **validation** L 2 M 2 N 3, simplicity M before L.

Choose M under the declared tie policy; do not use test scores to break the tie. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Final testing and honest claims. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/25, ai-4/27, ai-4/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Train, Validation, Test lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Select a model using a declared **validation** rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Where should parameter **fitting**, setting selection and final reporting occur?

Supported practice answer

Reasoned practice answer: Teacher explanation: Fit on **training**, select on **validation**,

evaluate the frozen choice on held-out test; fit pre**process**ing only on **training**.

Check every learner before discussion: Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: **validation** MAE 3 ties C, and A wins the simplicity tie. B has **validation** MAE 5 despite **training** MAE 0.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Require a **prediction** and a reason before comparing. Expected result: Select A using the predeclared simpler-model tie-break.

Reason: The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

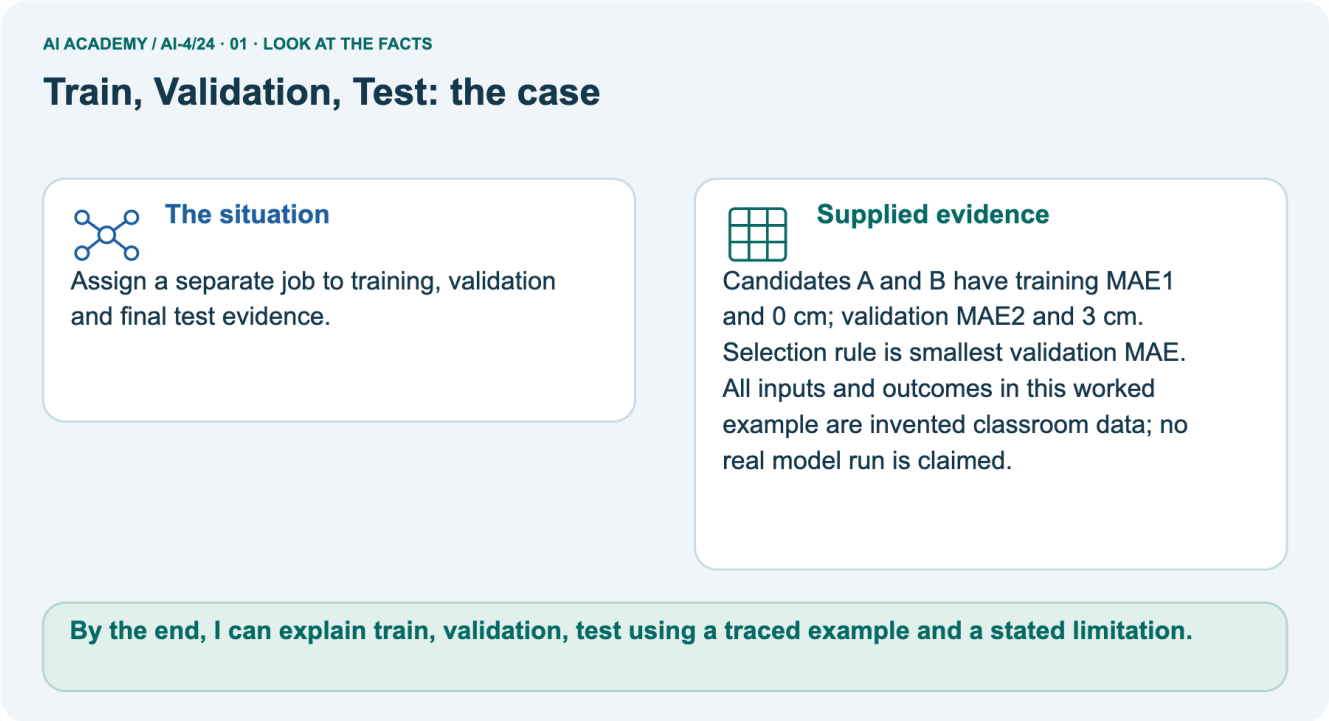
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Train, Validation, Test graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Train, Validation, Test: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Assign a separate job to **training**, **validation** and final test **evidence**. Supplied facts: Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

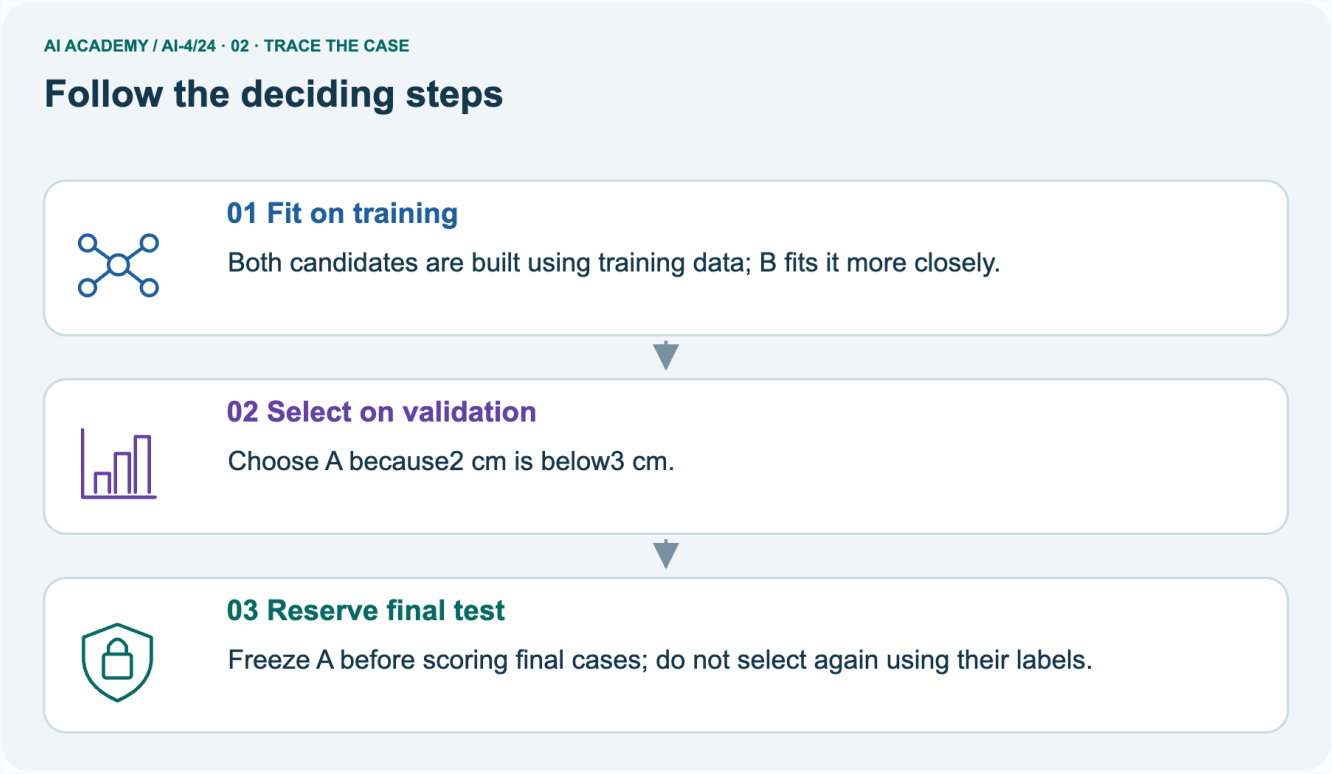
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

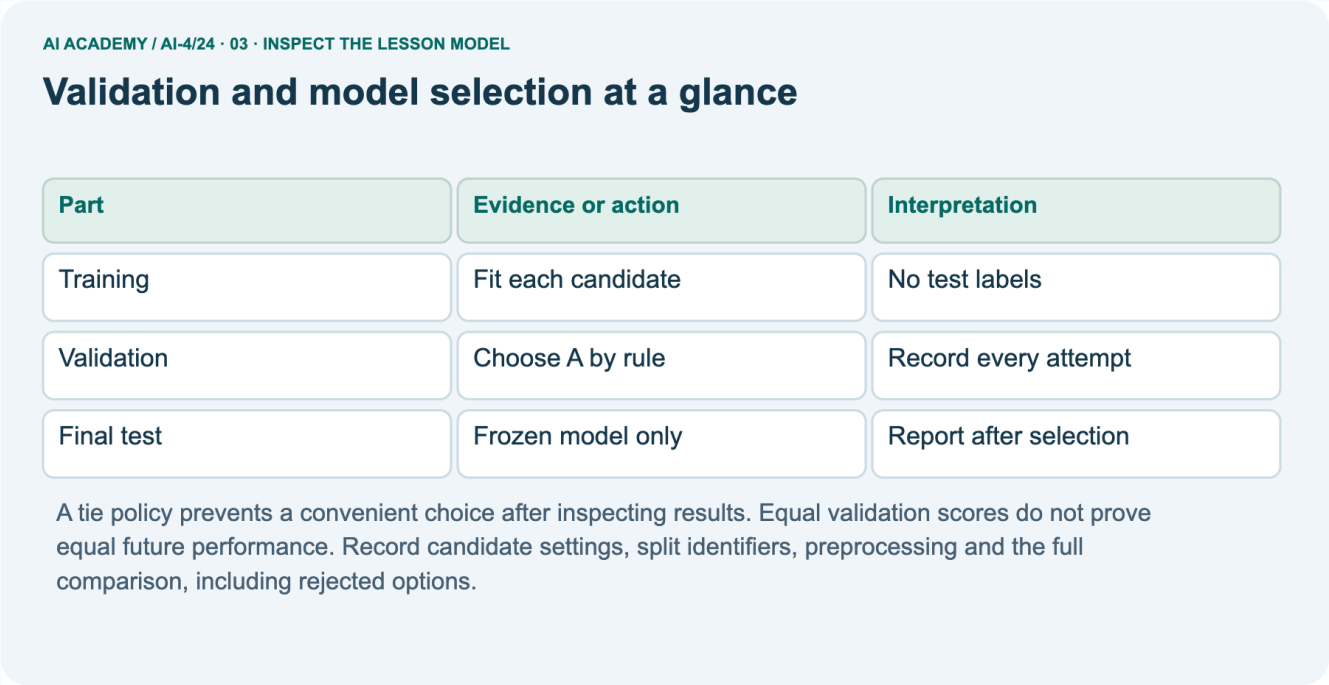
Fit on **training**: Both candidates are built using **training** data; B fits it more closely. → Select on **validation**: Choose A because2 cm is below3 cm. → Reserve final test: Freeze A before scoring final cases; do not select again using their labels.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Validation and model selection at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

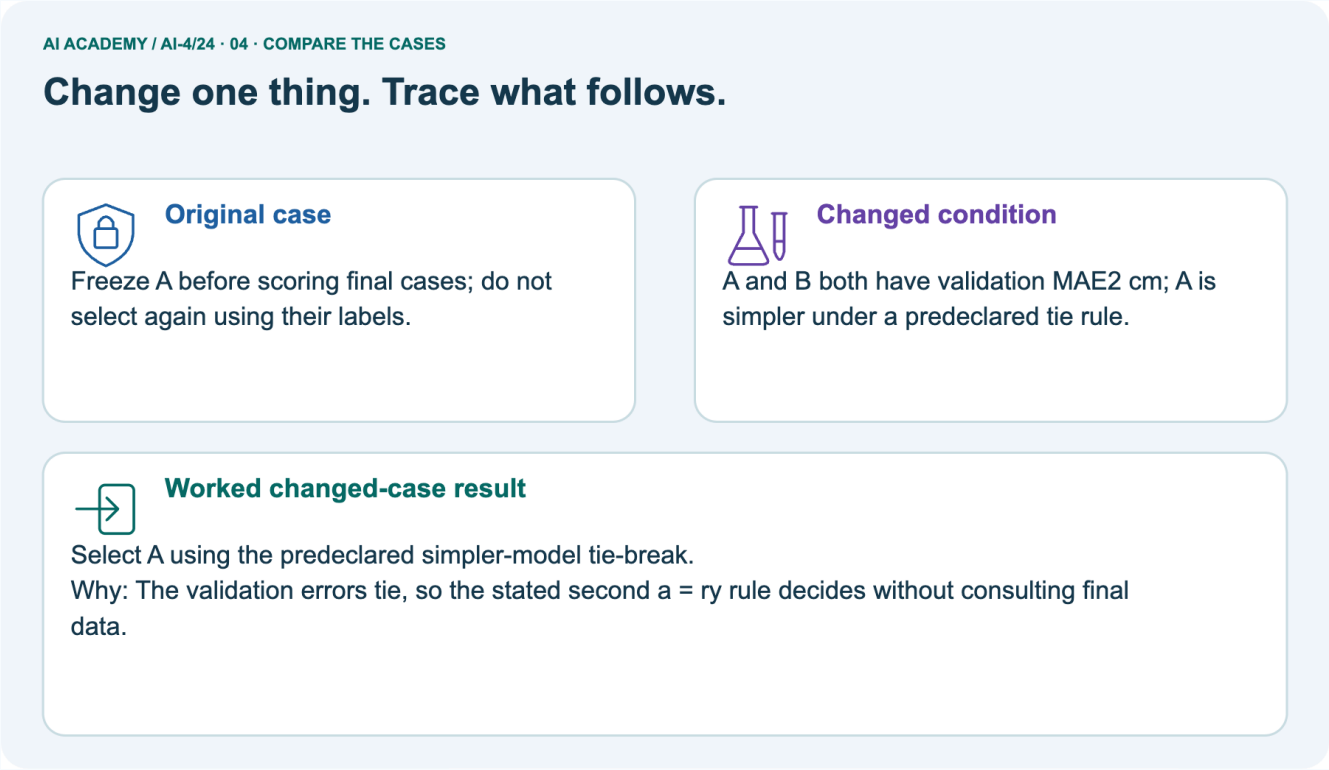
Validation and model selection at a glance. **Training** | Fit each candidate | No test labels. **Validation** | Choose A by rule | Record every attempt. Final test | Frozen model only | Report after selection

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Freeze A before scoring final cases; do not select again using their labels. Changed condition: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Expected result: Select A using the predeclared simpler-model tie-break. Reason: The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

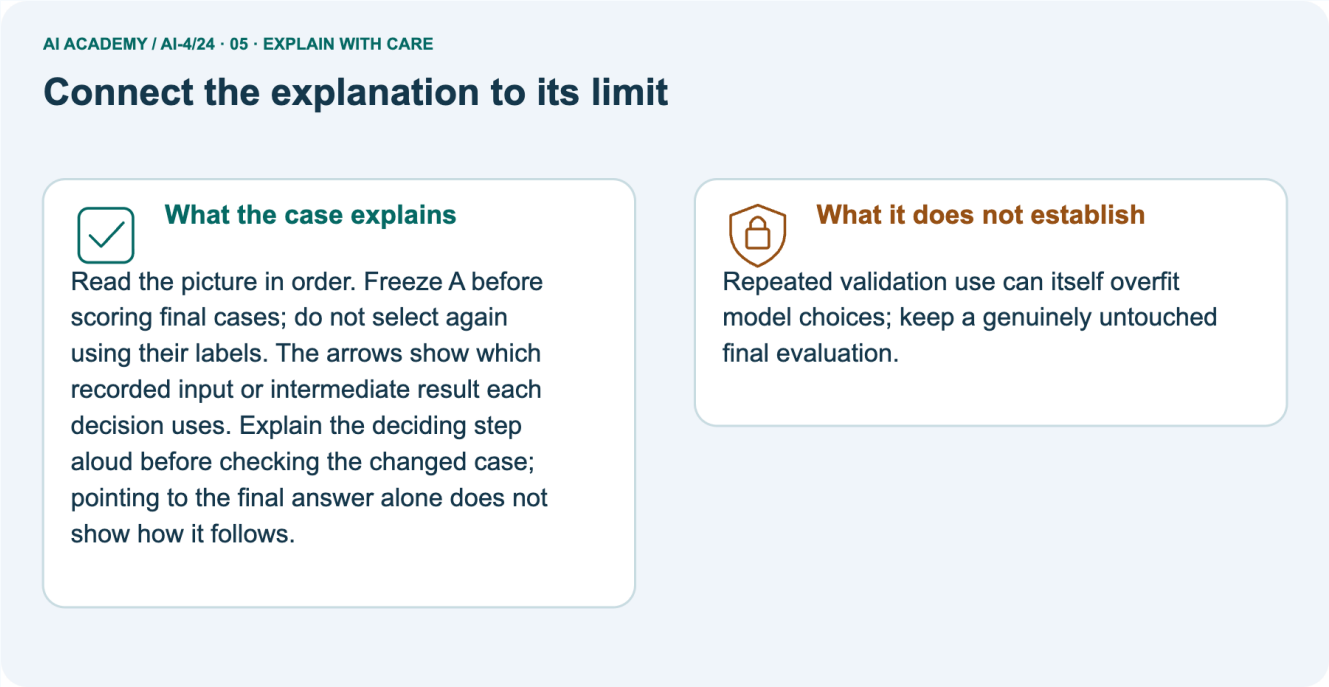
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

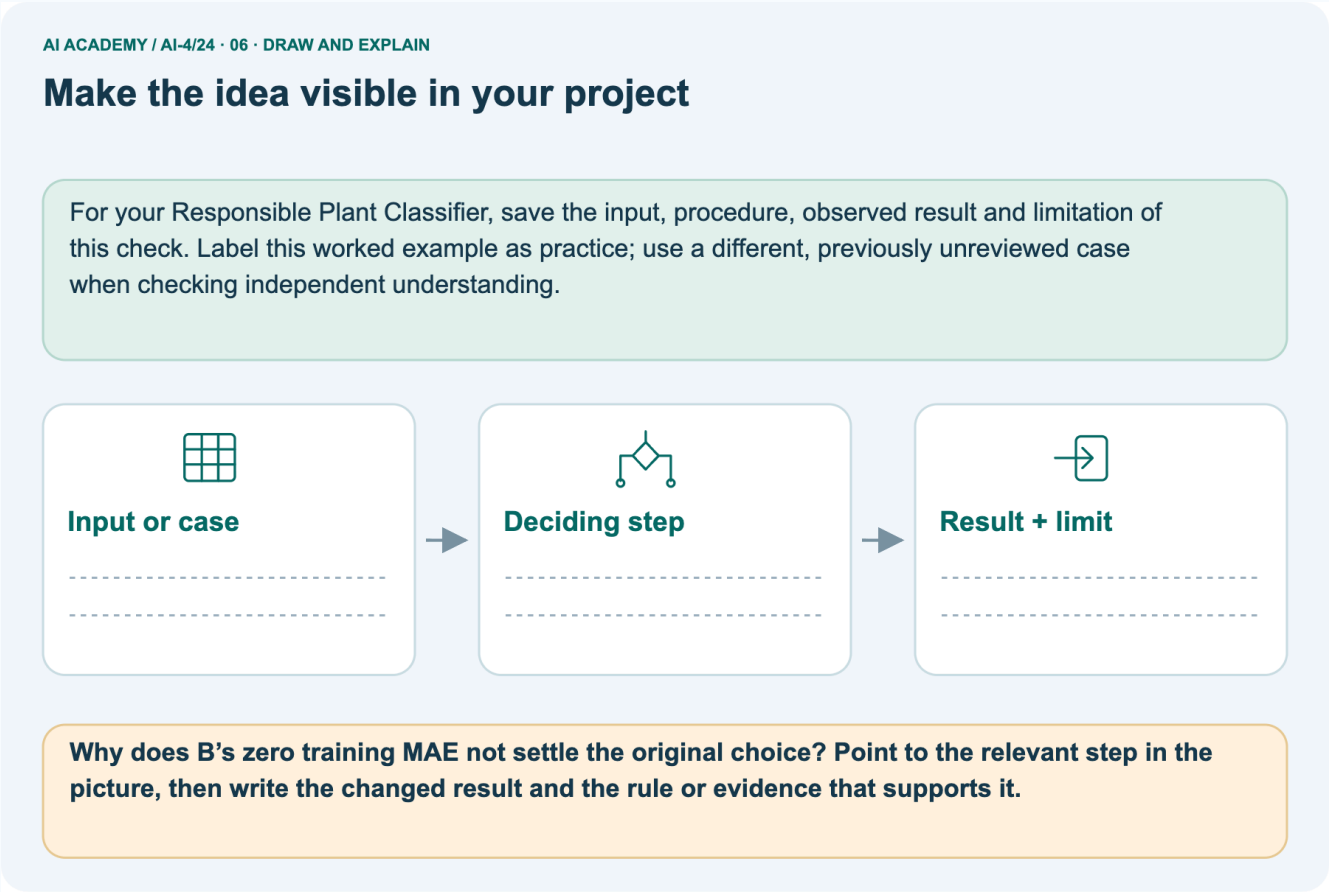
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Model selection is based on held-out **validation** error here. Lower **training** error can reflect **fitting** the **training** details rather than better new-case **prediction**.

Changed-case check: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated second a = ry rule decides without consulting final data.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.